QA AUTOMATION · AI-ASSISTED TESTING

AI-Driven Test Script Generation

Robot Framework · High-Level Approach

Version 1.0 · November 2026 · Internal

Executive summary

AIDLC takes us from a requirement to test cases, but it does not create Robot Framework test scripts. **We have built AI skills that close this gap:** they first learn our existing test automation, then turn AIDLC test cases (and, next, qTest test cases) into working, verified Robot Framework scripts that are added to the existing regression suite only after user approval.

Key principles

01 Repository knowledge
Builds a verified knowledge base of the existing tests, keywords and locators before any script is generated.

02 Existing keywords first
Generates scripts from existing keywords and locators; creates new ones only for new screens or features.

03 Verified and approved
Every script passes a full Robot Framework run and is added to the existing suite only after user approval.

1. Our approach

To write a reliable script for a new feature, the AI must first understand our existing automation, the same way a new tester would. The approach therefore has two steps.

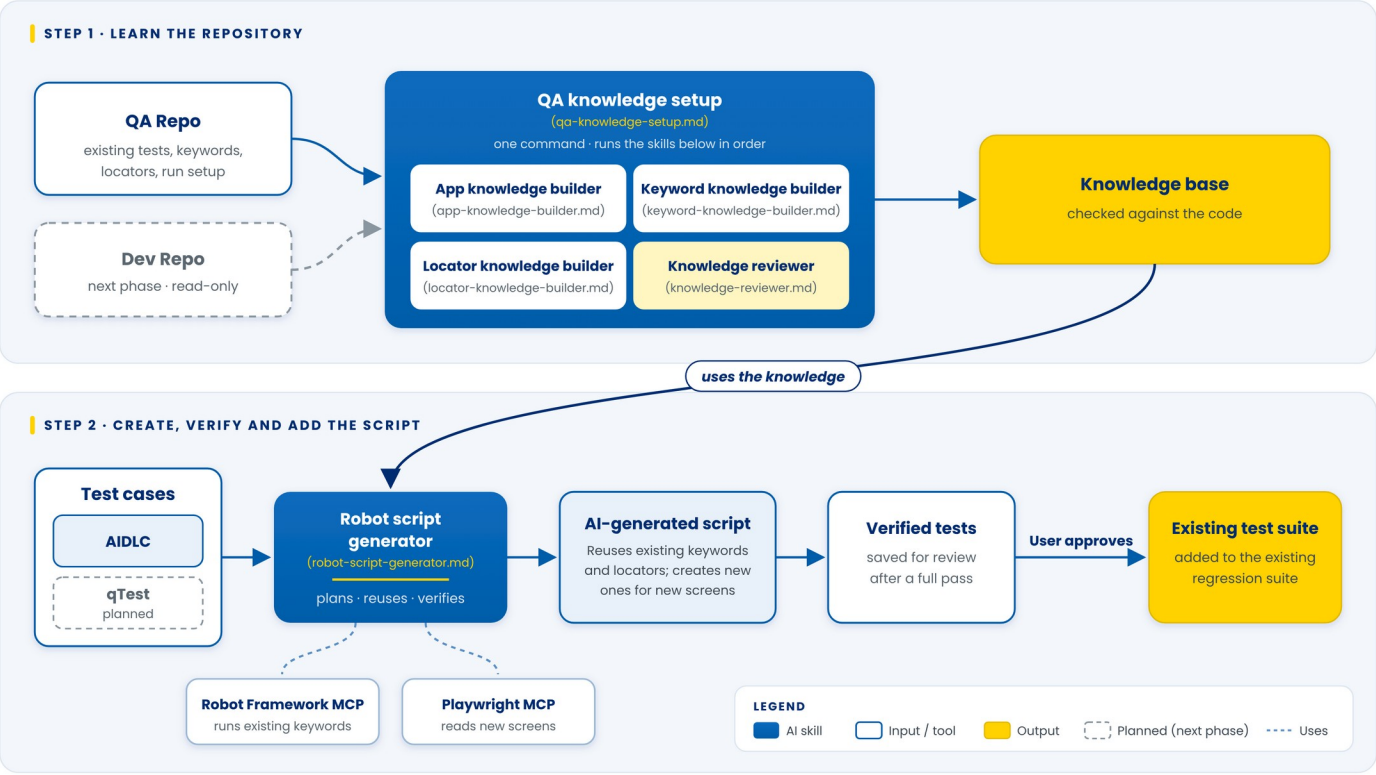


Figure 1. End-to-end flow: from repository knowledge to a verified script in the existing suite.

Step 1: Learn the repository (knowledge skills)

Skill	What it does
QA knowledge setup (qa-knowledge-setup.md)	The single command that runs all the skills below in the right order and builds the knowledge base.
App knowledge builder (app-knowledge-builder.md)	Learns the application: screens, business journeys, an application map showing how screens connect, business terms, and how tests are run.
Keyword knowledge builder (keyword-knowledge-builder.md)	Lists the existing keywords, what each one does and which business step it covers.
Locator knowledge builder (locator-knowledge-builder.md)	Lists where each button and field sits on screen, and how reliable each reference is.
Knowledge reviewer (knowledge-reviewer.md)	Checks every entry against the real code before the knowledge is trusted.

Knowledge can be built for a single suite, several suites, or the whole repository, and is refreshed when tests change.

Step 2: Create, verify and add the script (Robot script generator)

1. Reads the test case (AIDLC today; qTest planned) and the knowledge base.
2. Plans the test: which steps already exist as keywords, and which are new.
3. Uses **Robot Framework MCP** to drive the live application with existing keywords and reach the right screen.
4. Uses **Playwright MCP** to read the new screen and identify new fields and buttons.
5. Produces the **AI-generated script**: existing keywords and locators are reused; new ones are created only for new screens or features.
6. Verifies every new element, runs the full test, and saves it to the **verified tests** area only after a complete pass.
7. **A user reviews and approves** the test before it is added to the existing regression suite.

2. Where this fits

1. **Requirement:** business requirements and design documents (for example, the HLD).
2. **Test plan:** created with AIDLC from the requirement.
3. **Test cases / scenarios:** created with AIDLC from the test plan today; qTest test cases are planned as a second source.
4. **Test scripts (our skills):** Robot Framework scripts created from those test cases. AIDLC does not cover this step; this is the capability we built.

3. Controls and safeguards

- Existing keywords and locators are always reused; new ones are created only for new screens or features.
- Every new screen element is verified on a running screen before it is used.
- A test is reported as passed only after a complete Robot Framework run.
- The AI never changes existing tests, keywords or locators; it recommends fixes for user approval.
- Nothing is added to the regression suite or committed without user approval.
- Credentials come only from the existing secure vault and are never displayed or stored.

4. Progress checklist

Overall: 3 of 7 complete · 1 in progress · 3 pending

#	Item	What it covers	Status
1	**Knowledge skills**	Application, keyword and locator knowledge built and reviewed, with one command to build or refresh it.	**Done**
2	**Robot script generator**	Generates, verifies and runs Robot Framework scripts from AIDLC test cases; working end to end in UAT.	**Done**
3	**Mock screen support**	Generates scripts for unreleased features from mock screens; the final test is held until release.	**Done**
4	**Development repository knowledge**	Read-only understanding of the development code: screens, fields, validation rules and access by role.	**In progress**
5	**Scripts from development knowledge**	Generates scripts for new screens directly from development knowledge and existing keywords, without needing a mock screen.	**Pending**
6	**qTest test case input**	Accepts qTest test cases as a second source, alongside AIDLC.	**Pending**
7	**Central access**	One shared, version-controlled copy of the skills for all QA users in Devin and VS Code.	**Pending**

END GOAL

From requirement to Robot Framework test script: every new requirement flows from AIDLC test cases to a verified, user-approved Robot Framework script in the existing regression suite, built on existing keywords and locators with minimal manual effort.